

```
!pip install langchain
```

```
Requirement already satisfied: langchain in /usr/local/lib/python3.12/dist-packages (0.3.27)
Requirement already satisfied: langchain-core<1.0.0,>=0.3.72 in /usr/local/lib/python3.12/dist-packages (from langchain) (0.3.78)
Requirement already satisfied: langchain-text-splitters<1.0.0,>=0.3.9 in /usr/local/lib/python3.12/dist-packages (from langchain) (0.3.10)
Requirement already satisfied: langsmith>=0.1.17 in /usr/local/lib/python3.12/dist-packages (from langchain) (0.4.33)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain) (2.11.10)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in /usr/local/lib/python3.12/dist-packages (from langchain) (2.0.43)
Requirement already satisfied: requests<3,>=2 in /usr/local/lib/python3.12/dist-packages (from langchain) (2.32.4)
Requirement already satisfied: PyYAML>=5.3 in /usr/local/lib/python3.12/dist-packages (from langchain) (6.0.3)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<1.0.0,>=0.3.72) (9.0.0)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<1.0.0,>=0.3.72) (1.33.0)
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<1.0.0,>=0.3.72) (4.12.0)
Requirement already satisfied: packaging<26.0.0,>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<1.0.0,>=0.3.72) (25.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith>=0.1.17->langchain) (0.27.0)
Requirement already satisfied: orjson>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith>=0.1.17->langchain) (3.11.0)
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith>=0.1.17->langchain) (1.0.0)
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Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2->langchain) (3.4.0)
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2->langchain) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2->langchain) (2025.11.12)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->langsmith>=0.1.17->langchain) (0.14.0)
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->langchain) (3.0.0)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.12/dist-packages (from anyio->httpx<1,>=0.23.0->langsmith>=0.1.17->langchain) (1.3.1)
```

```
!pip install -q google-generativeai langchain
```

```
import google.generativeai as genai
import os
```

```
os.environ["GOOGLE_API_KEY"] = "AIzaSyDvmXzyjXOHipKFddCzIUxpCWBFIb68fH0"
```

```
genai.configure(api_key=os.environ["GOOGLE_API_KEY"])
```

```
model = genai.GenerativeModel("gemini-2.5-flash")
```

```
import json
```

```
def semantic_plan(goal: str):
    prompt = f"""
    You are an intelligent task planner. Decompose the following goal into a clear, ordered list of subtasks.
    Each subtask should be actionable, specific, and logically sequenced.
```

```
Goal: {goal}
```

```
Return the plan in numbered steps.
"""
    response = model.generate_content(prompt)
    return response.text.strip()
```

```
def parse_plan(plan_text: str):
    tasks = []
    for line in plan_text.split("\n"):
        if line.strip() and line[0].isdigit():
            tasks.append(line[line.find('.')+1:].strip())
    return tasks
```

```
class TaskExecutor:
    def __init__(self):
        self.context = {}

    def execute(self, task):
        print(f"🚀 Executing: {task}")
        # Placeholder execution logic
        result = f"Completed: {task}"
        self.context[task] = result
        return result
```

```
def run_semantic_planner(goal: str):
    display(Markdown(f"### 🎯 **Goal:** {goal}"))
    print("_____")

    # Generate plan using Gemini
    plan_text = semantic_plan(goal)
    display(Markdown(f"#### 📋 **Generated Plan:**"))
    display(Markdown(f"```\n{plan_text}\n`"))

    # Parse into structured list
    tasks = parse_plan(plan_text)
    executor = TaskExecutor()

    print("\n_____")
    display(Markdown(f"#### 🚀 **Executing Tasks Sequentially:**"))
    print("_____")
```

```
for i, task in enumerate(tasks, start=1):
    print(f"Step {i:<2} | 🌟 {task}")
    result = executor.execute(task)
    print(f"      ↳ ✅ {result}\n")

print("_____")
display(Markdown("### ✅ **All Tasks Completed Successfully!**"))
```

run_semantic_planner("Prepare a weekly marketing report and send it to the manager")

 Goal: Prepare a weekly marketing report and send it to the manager

 Generated Plan:

Here's a clear, ordered list of subtasks to prepare a weekly marketing report and send it to the manager:

1. ****Confirm Report Scope & Metrics:**** Review the standard requirements for the weekly marketing report, including key performance
2. ****Access Data Sources:**** Log in to all necessary marketing platforms (e.g., Google Analytics, social media insights, ad platfor
3. ****Extract Raw Data:**** Download or gather the relevant data for the reporting period from each platform.
4. ****Consolidate & Organize Data:**** Combine data from various sources into a single, manageable format (e.g., a spreadsheet), ensu
5. ****Analyze Performance:**** Review the consolidated data to identify trends, successes, areas for improvement, and key insights re
6. ****Outline Report Structure:**** Create a draft outline for the report, including sections for key metrics, analysis, conclusions,
7. ****Draft Report Content:**** Write the narrative for each section of the report, incorporating the analyzed data, charts, graphs,
8. ****Create Visualizations:**** Design and insert relevant charts, graphs, and tables to visually represent the data and enhance rea
9. ****Review and Edit Report:**** Proofread the entire report for accuracy, clarity, grammar, spelling, and completeness. Ensure all
10. ****Format Report for Delivery:**** Convert the report into the required final format (e.g., PDF, Google Doc, Word document, or pre
11. ****Draft Email to Manager:**** Compose a concise email to the manager, including a brief summary of the report's key findings or a
12. ****Attach Report (if applicable):**** Attach the final report file to the email.
13. ****Send Report:**** Send the email containing the report to the manager.

 Executing Tasks Sequentially:

- Step 1 | 🌟 ****Confirm Report Scope & Metrics:**** Review the standard requirements for the weekly marketing report, including key p
🌟 Executing: ****Confirm Report Scope & Metrics:**** Review the standard requirements for the weekly marketing report, including key
↳ ✅ Completed: ****Confirm Report Scope & Metrics:**** Review the standard requirements for the weekly marketing report, incl
- Step 2 | 🌟 ****Access Data Sources:**** Log in to all necessary marketing platforms (e.g., Google Analytics, social media insights,
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- Step 3 | 🌟 ****Extract Raw Data:**** Download or gather the relevant data for the reporting period from each platform.
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- Step 4 | 🌟 ****Consolidate & Organize Data:**** Combine data from various sources into a single, manageable format (e.g., a spreadsh
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- Step 5 | 🌟 ****Analyze Performance:**** Review the consolidated data to identify trends, successes, areas for improvement, and key i
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↳ ✅ Completed: ****Analyze Performance:**** Review the consolidated data to identify trends, successes, areas for improvement
- Step 6 | 🌟 ****Outline Report Structure:**** Create a draft outline for the report, including sections for key metrics, analysis, co
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↳ ✅ Completed: ****Outline Report Structure:**** Create a draft outline for the report, including sections for key metrics, a
- Step 7 | 🌟 ****Draft Report Content:**** Write the narrative for each section of the report, incorporating the analyzed data, charts
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↳ ✅ Completed: ****Draft Report Content:**** Write the narrative for each section of the report, incorporating the analyzed d
- Step 8 | 🌟 ****Create Visualizations:**** Design and insert relevant charts, graphs, and tables to visually represent the data and e
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- Step 9 | 🌟 ****Review and Edit Report:**** Proofread the entire report for accuracy, clarity, grammar, spelling, and completeness. E
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↳ ✅ Completed: ****Review and Edit Report:**** Proofread the entire report for accuracy, clarity, grammar, spelling, and comp
- Step 10 | 🌟 ****Format Report for Delivery:**** Convert the report into the required final format (e.g., PDF, Google Doc, Word docum
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↳ ✅ Completed: ****Format Report for Delivery:**** Convert the report into the required final format (e.g., PDF, Google Doc,
- Step 11 | 🌟 ****Draft Email to Manager:**** Compose a concise email to the manager, including a brief summary of the report's key fin
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- Step 12 | 🌟 ****Attach Report (if applicable):**** Attach the final report file to the email.
🌟 Executing: ****Attach Report (if applicable):**** Attach the final report file to the email.
↳ ✅ Completed: ****Attach Report (if applicable):**** Attach the final report file to the email.
- Step 13 | 🌟 ****Send Report:**** Send the email containing the report to the manager.
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↳ ✅ Completed: ****Send Report:**** Send the email containing the report to the manager.

✅ **All Tasks Completed Successfully!**

Start coding or [generate](#) with AI.

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